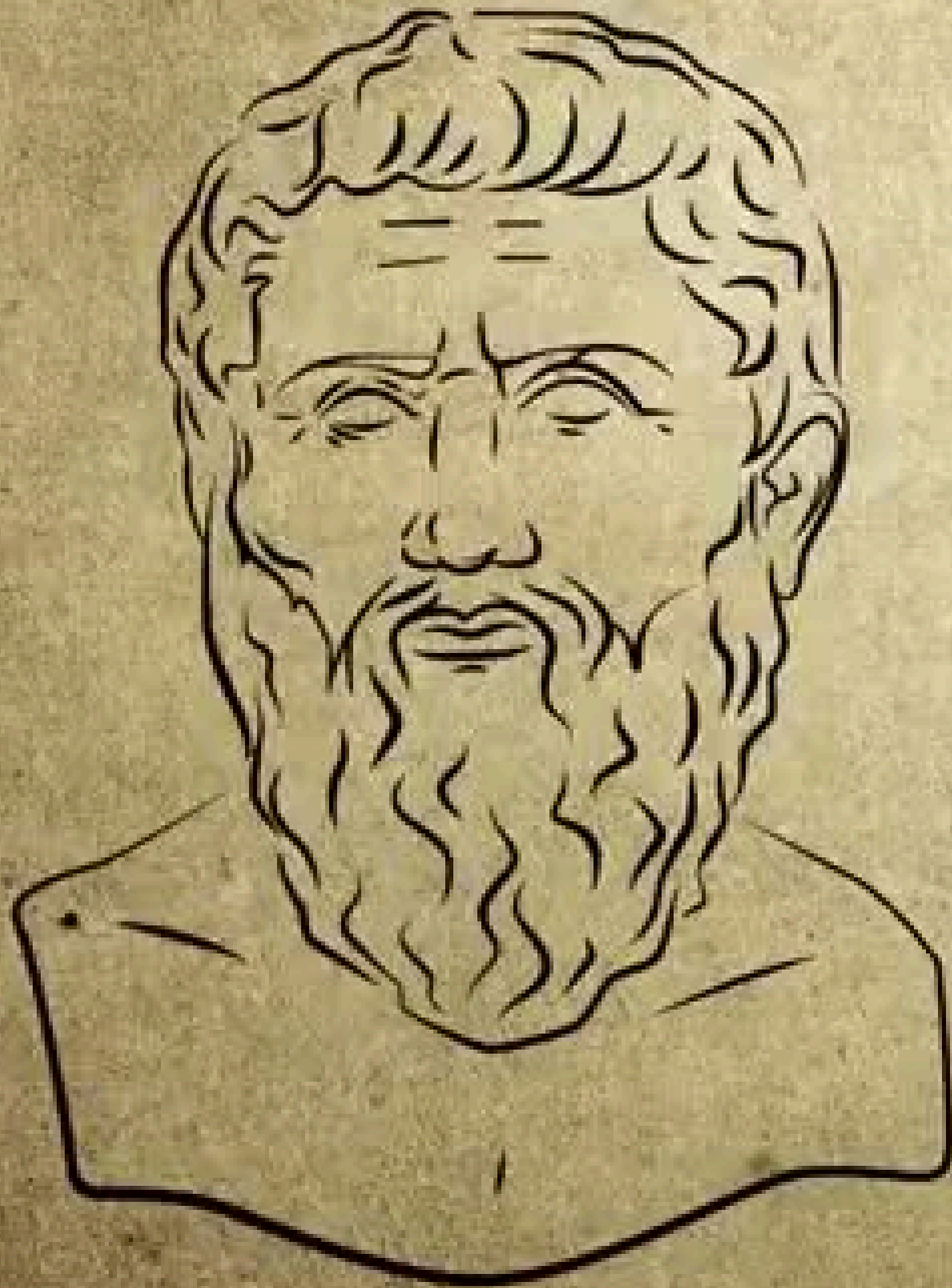


NETWORKS AND THE EFFECTS OF USING THEM

LEARNING OBJECTIVES

NETWORKS: Routers, Common Network Devices, Wifi and Bluetooth, Cloud Computing, Intranets, extranets and the Internet, LAN, WLAN and WAN's

Network Issues and Communication: Security, Anti Malware and Electronic Conferencing



*The beginning is
the most
important part of
the work*

Plate

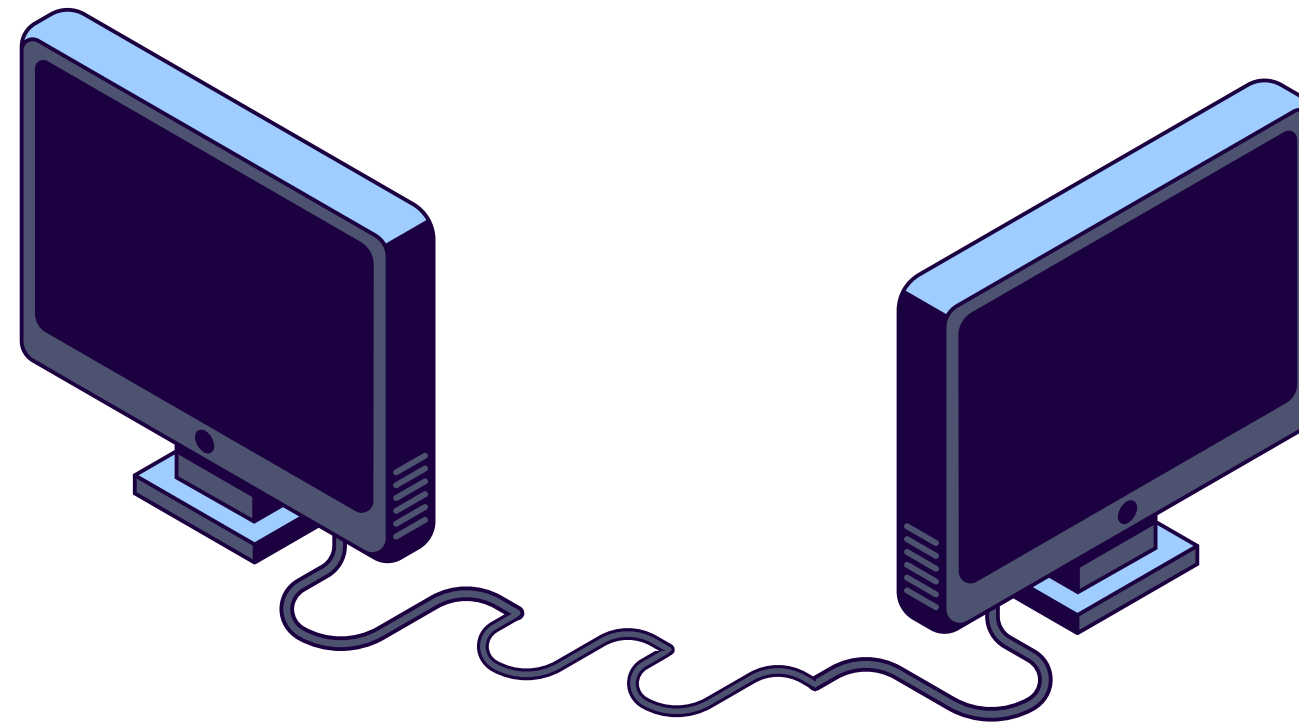
NETWORKS

Routers, Switches, and Wi-Fi: The Building Blocks of Networks



WHAT IS A NETWORKS?

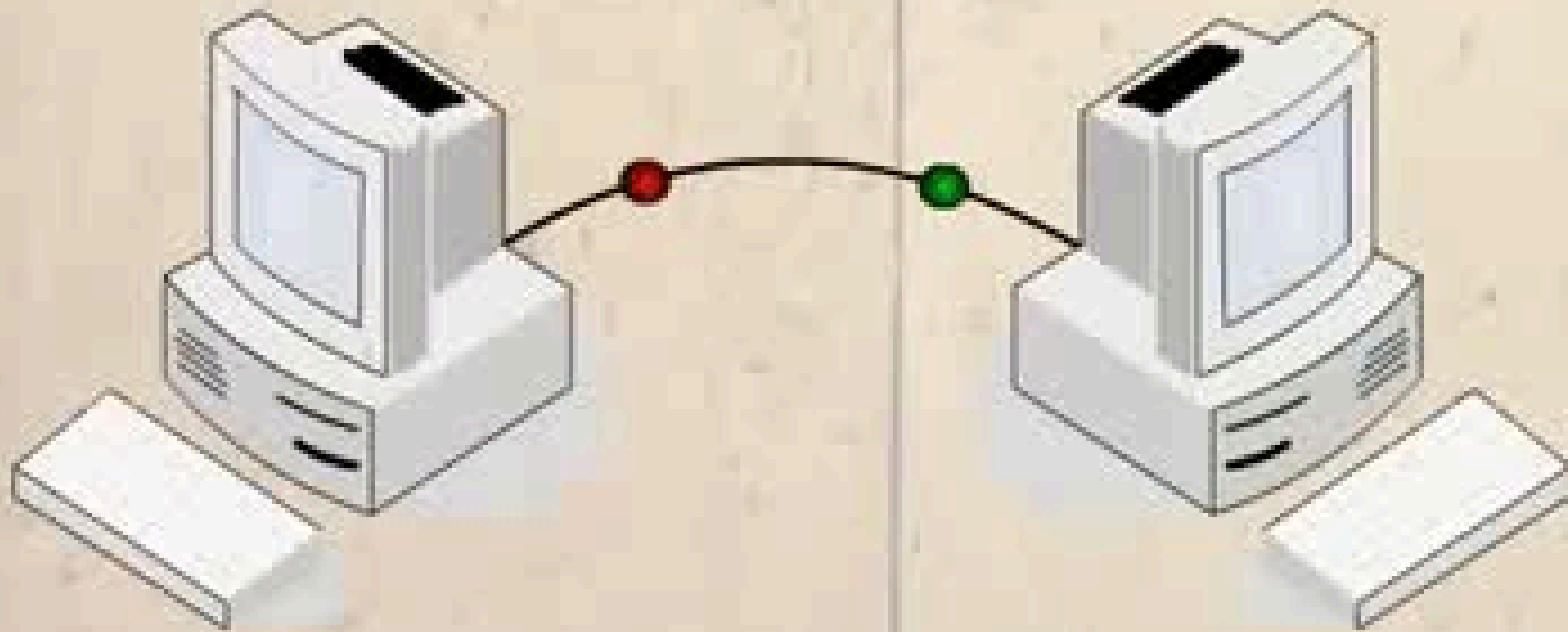
It is a group of connected devices, like computers, phones, or servers, that can communicate with each other to share information and resources.



These connections can be wired, like with cables, or wireless, using signals like Wi-Fi.

What is a Network?

A network is a group of devices that are connected in such a manner that they can share information and resources with each other



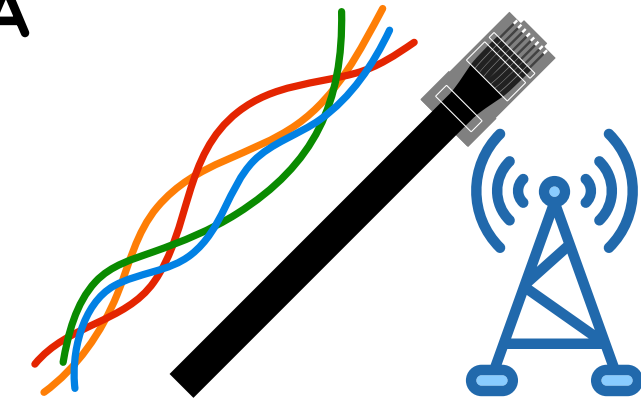
4 COMPONENTS OF NETWORK

NETWORK DEVICES

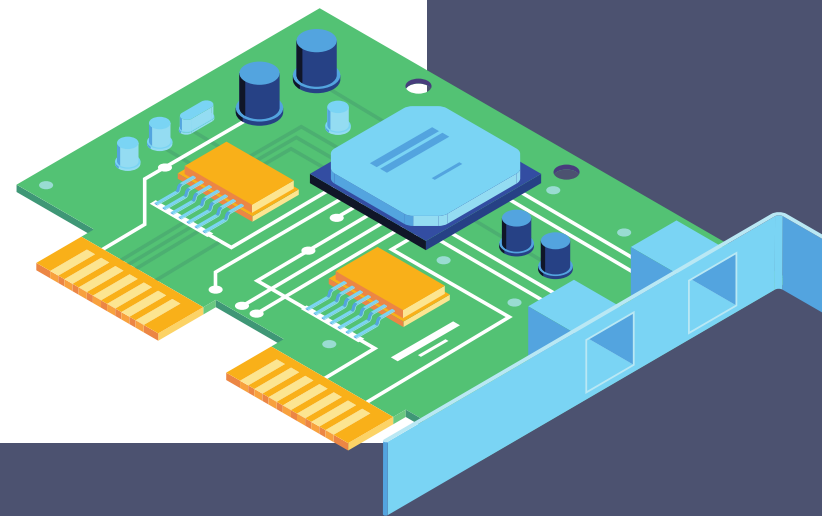
It is called Hosts



NETWORK MEDIA



NETWORK INTERFACE CARD (NIC)



NETWORK PROTOCOL

HTTPS
DNS



NETWORK DEVICES

- is any hardware that helps computers, phones, and other gadgets connect and communicate with each other in a network.
- These devices can send, receive, and direct data so it reaches the right place.
- Examples include **routers** (send data between networks), **switches** (connect devices within the same network), and access points (**provide Wi-Fi**).

NETWORK DEVICES



NIC (Network Interface Card)

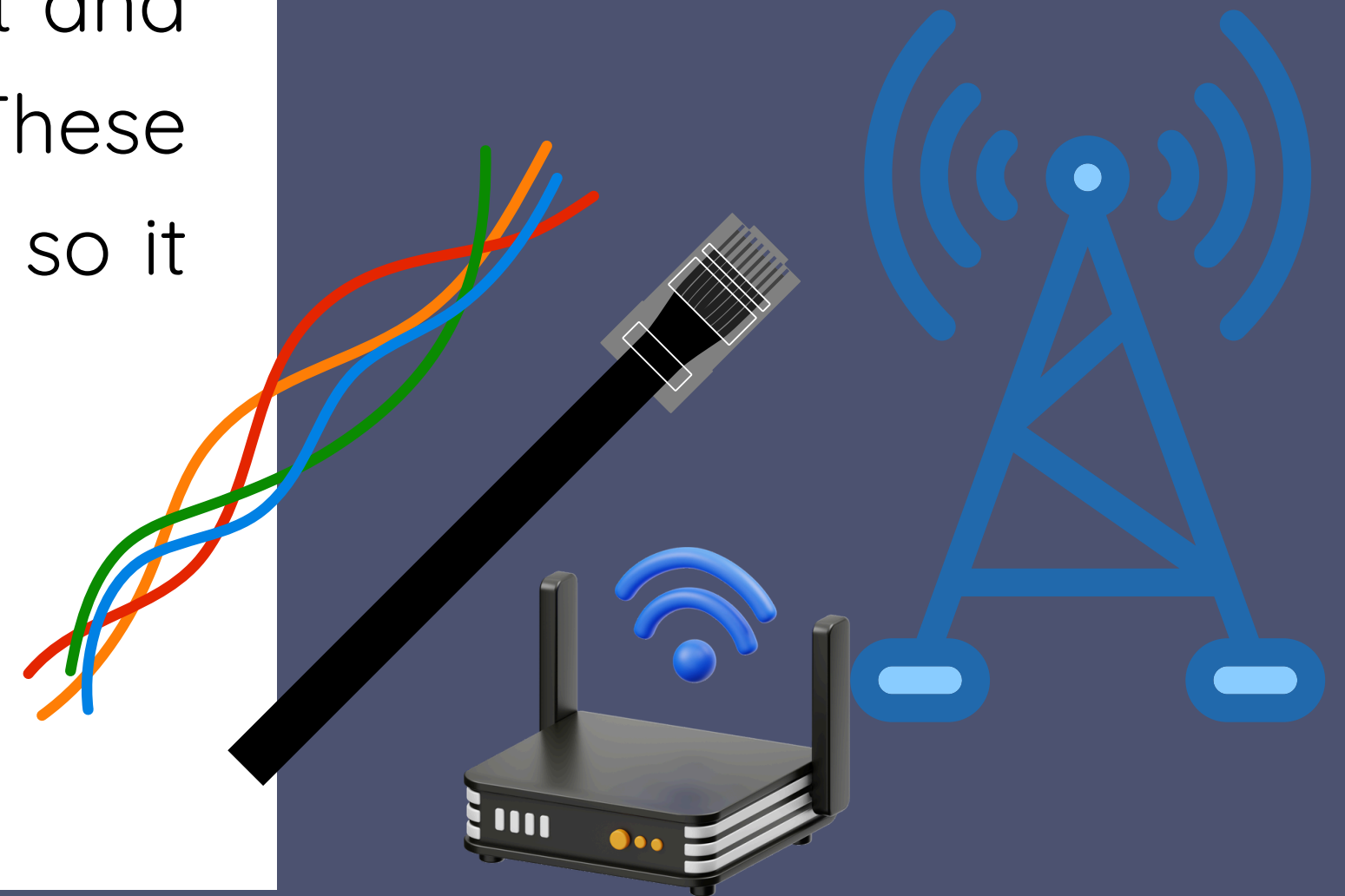
- It is a hardware component that allows a computer or device to connect to a network. It acts as a bridge between the device and the network, enabling communication with other devices.



**NETWORK
INTERFACE
CARD**

NETWORK MEDIA

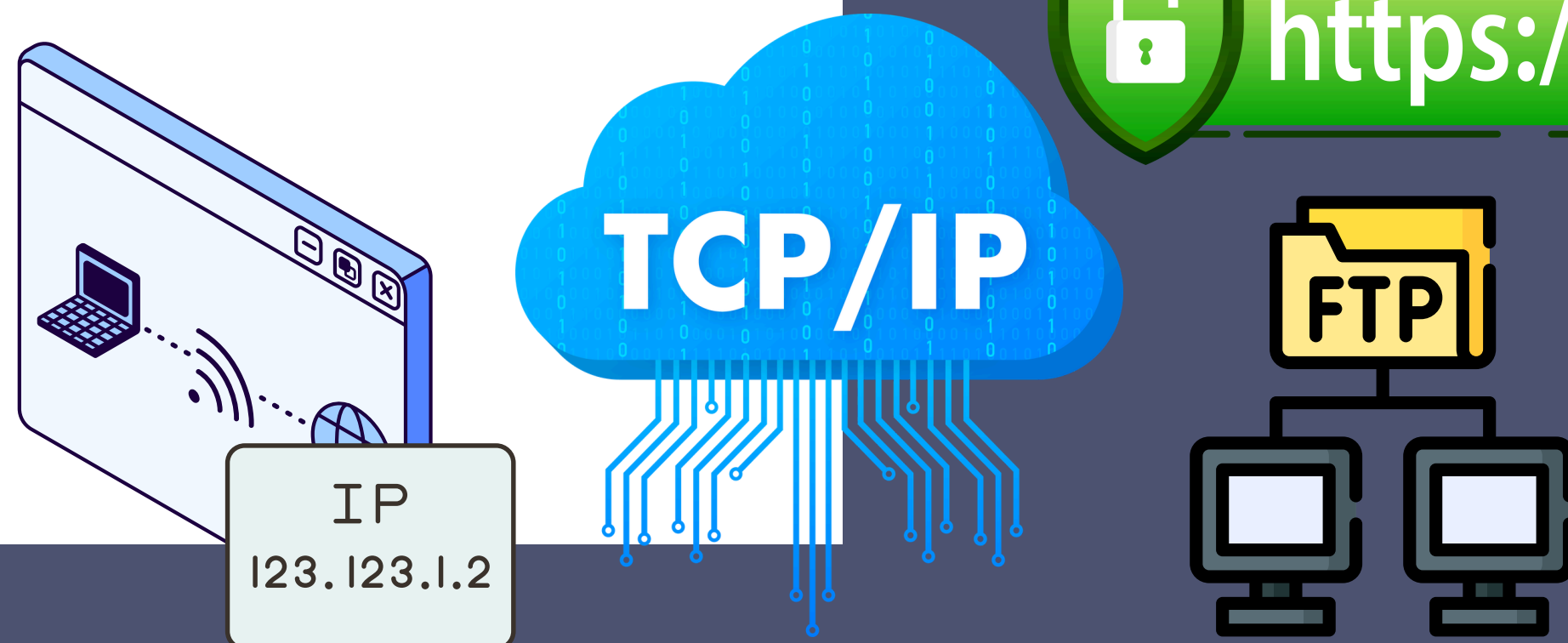
- is the physical path that data travels through in a network.
- It is a piece of equipment that helps computers and other gadgets connect and communicate with each other. These devices send, receive, and direct data so it reaches the right destination.



NETWORK MEDIA

Network Protocol

- is a set of rules that devices use to communicate with each other over a network.
- It tells devices how to send, receive, and understand data so that information travels correctly from one device to another.



NETWORK PROTOCOL

ANALOGY EXAMPLE

Imagine you are sending a letter:



Network Device = the person sending the letter



Network NIC = the person's ability to send/receive



Network Media = the road used to deliver it



Protocol = the rules for how the letter is addressed and delivered

MAC (Media Access Control) Address

- It is a number which uniquely identifies a device when it is connected to a network.
- It is made up of 48 bits which are shown as 6 groups of hexadecimal digits with the general format:

NN - NN - NN - DD - DD - DD
Manufacturer's code Device serial number

MAC Address: 00 - 1C - B3 - 4F - 25 - FF

**MEDIA ACCESS
CONTROL
(MAC)
ADDRESS**

IP (Internet Protocol) Address

- It is a unique string of numbers that identifies a device on a network, like the internet.
- is a numerical label assigned to each device connected to a network that uses the Internet Protocol for communication.

**INTERNET
PROTOCOL (IP)
ADDRESS**

There are two version of Internet Protocol (IP)

IPv4:

192.168.1.1

IPv6

A8FB:7A88:FFFF:0FFF:3D21:2085

IPv4 Addresses (32-bit)

1. **192.168.1.1** - Commonly used as the default gateway for home networks.
2. **10.0.0.1** - Often used for private networks, like those in businesses.
3. **172.16.254.1** - Another example of a private network IP address.
4. **8.8.8.8** - Public IP address used by Google's DNS servers.
5. **127.0.0.1** - The loopback address, used to refer to the local computer.

IPv6 Addresses (128-bit)

1. **2001:0db8:85a3:0000:0000:8a2e:0370:7334** - Example of a typical IPv6 address.
2. **fe80::1ff:fe23:4567:890a** - A link-local address used in IPv6 for communication within a single network segment.
3. **2001:4860:4860::8888** - Google's public DNS server in IPv6 format.

IPv4

- It is the internet Protocol version 4.
- The older system with fewer addresses, and we're running out of them.
- Decimal numbers separated by dots

IPv6

- It is the internet Protocol version 6.
- It was created to provide a huge number of addresses and improve features like speed, security, and auto-configuration.
- Hexadecimal numbers separated by colons

Data Packets

- It is a small piece of information that is sent over a network.
- When you send or receive data online, like an email or a video, it's broken down into these smaller packets so it can be transmitted efficiently.

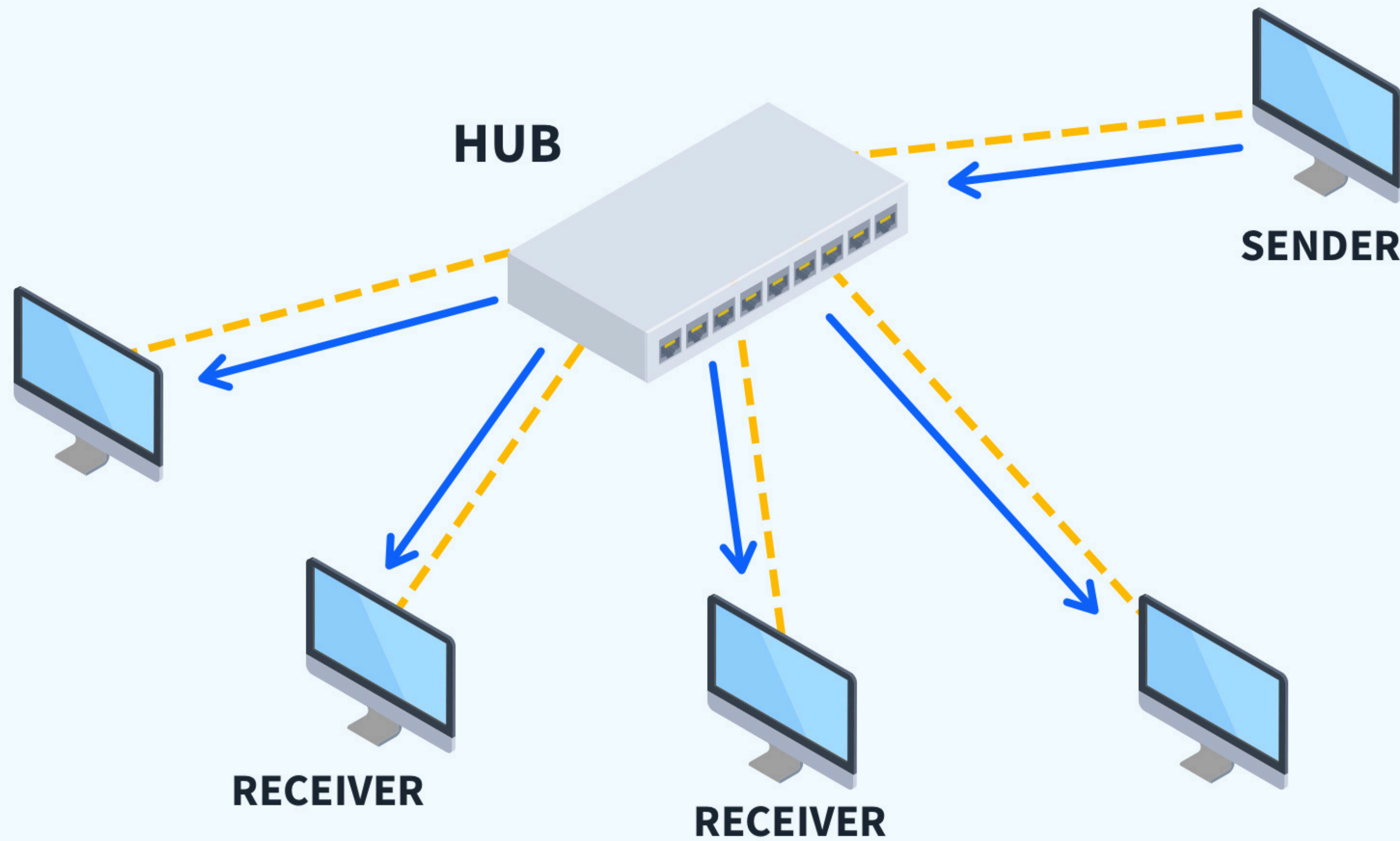
**DATA
PACKETS**

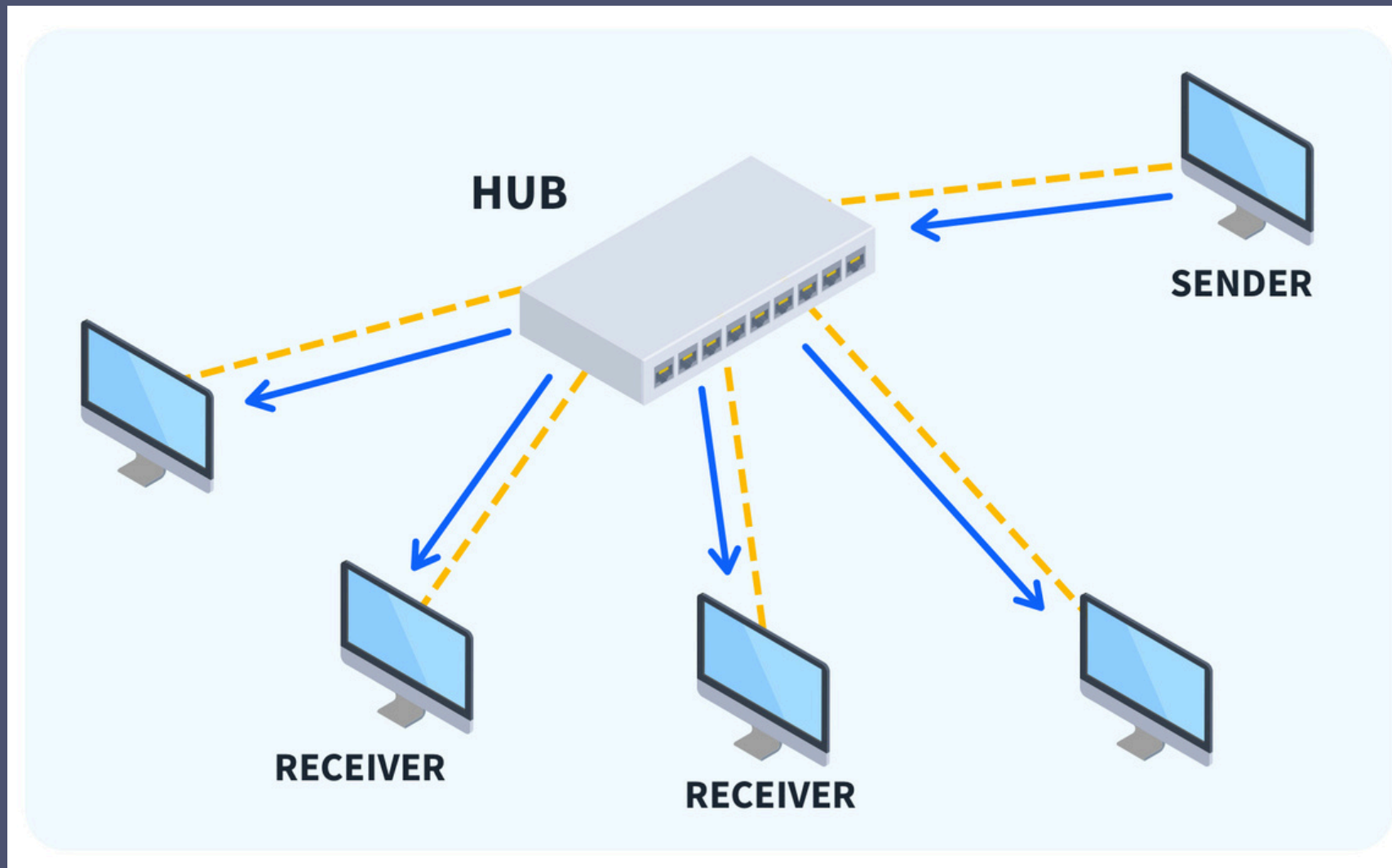
HUB

HUBS

- are hardware devices that can have a number of other devices connected to them.
- They are used primarily to connect devices together to form a local area network (LAN) often in the same building
- A basic networking device that broadcasts data to all devices on the network.
- Less efficient and suitable for simpler network setups.







HUB NETWORK SETUP

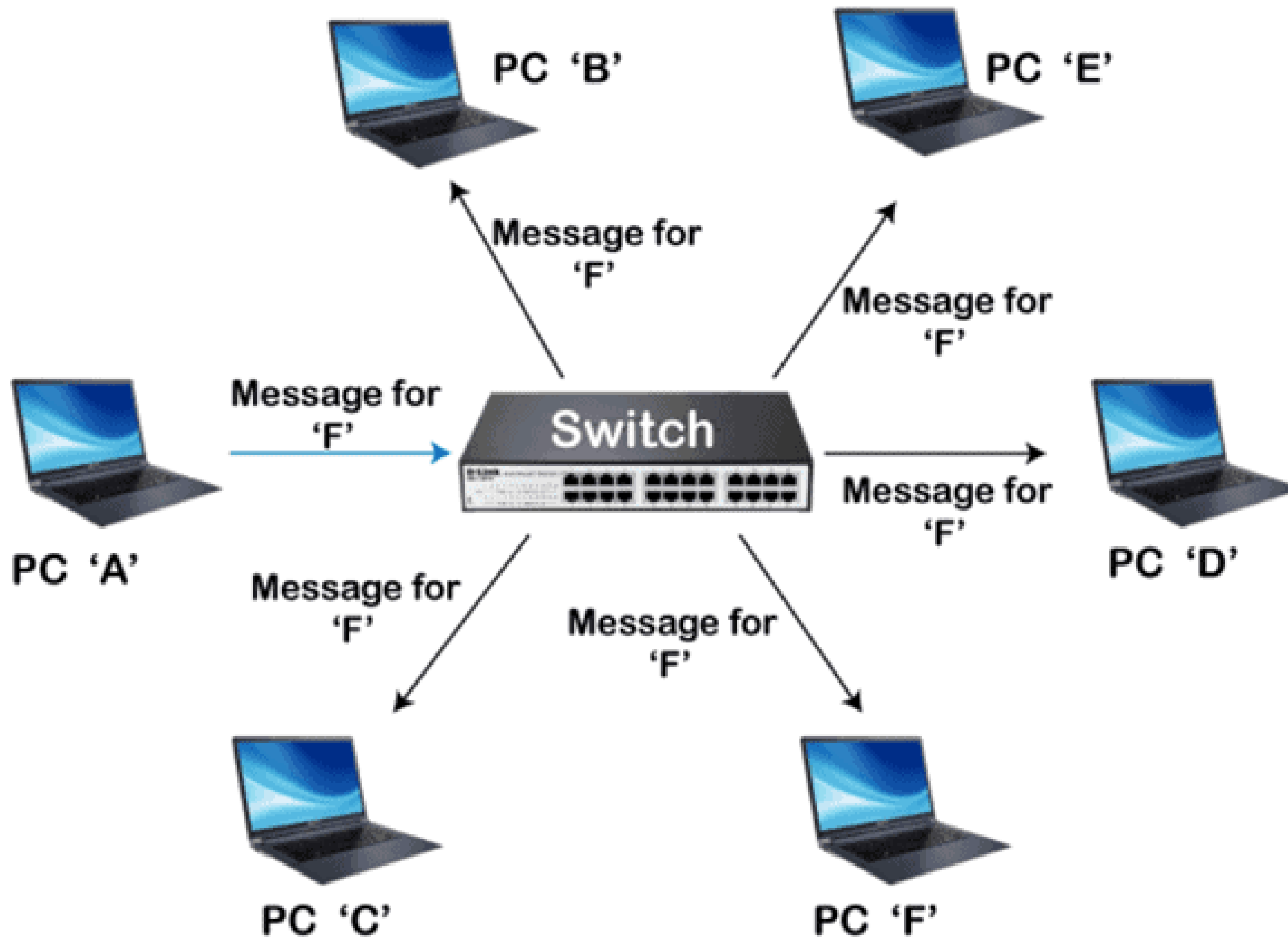
- The Hub receives data from the Sender computer.
- instead of sending it only to the intended recipient, the Hub broadcasts the data to all connected devices (Receivers).
- Every device receives the message, but only the intended recipient uses it, others just ignore it.
- This creates unnecessary network traffic and can lead to collisions.
- Not really a secure network when it comes to data transfer

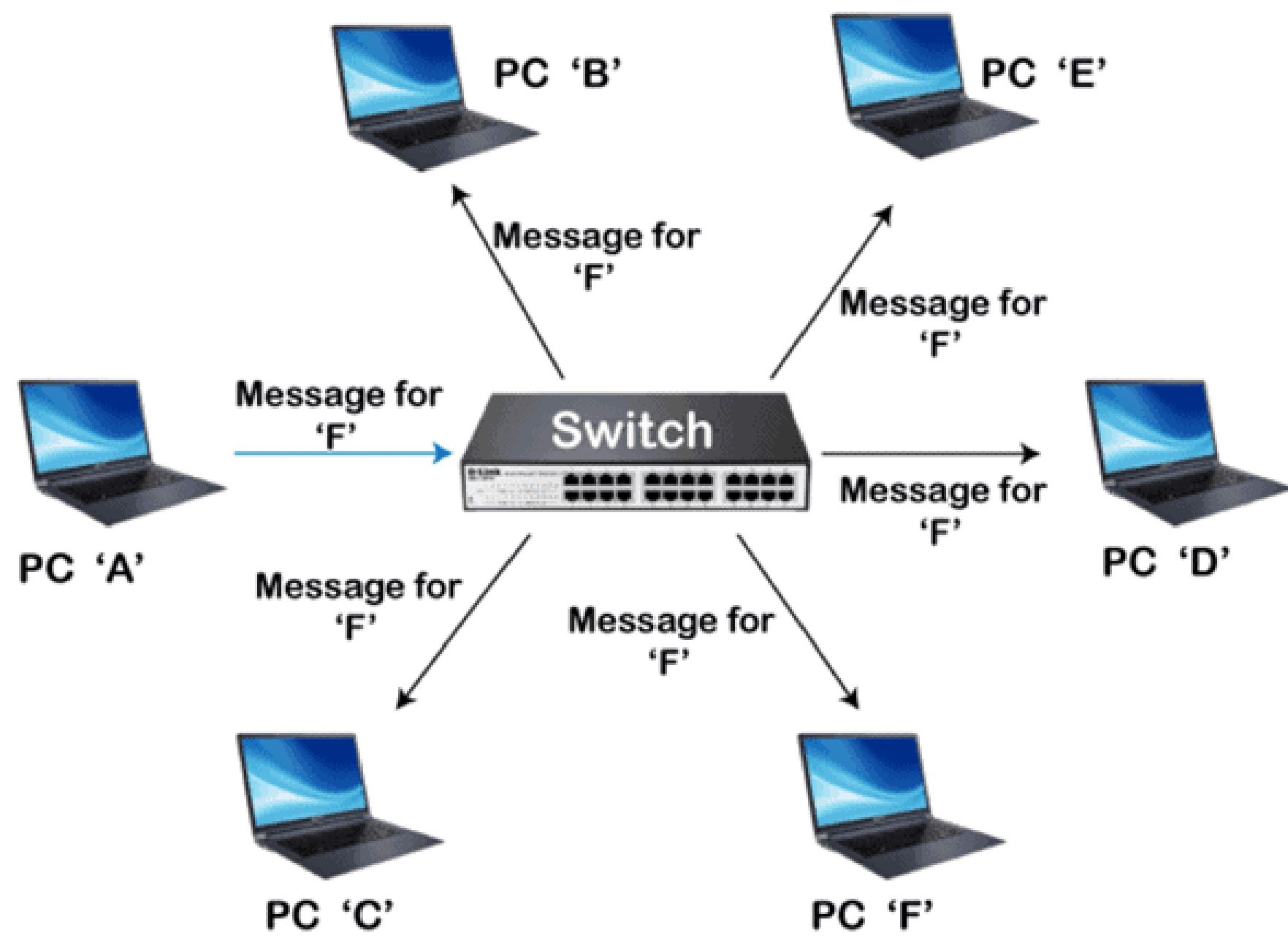
SWITCHES

SWITCHES

- are “Intelligent” version of Hub.
- A more advanced device that directs data only to the intended recipient using MAC addresses. More efficient and ideal for larger, more complex networks.
- More secure than Hub (because only the intended device is sent the data) and do not waste bandwidth

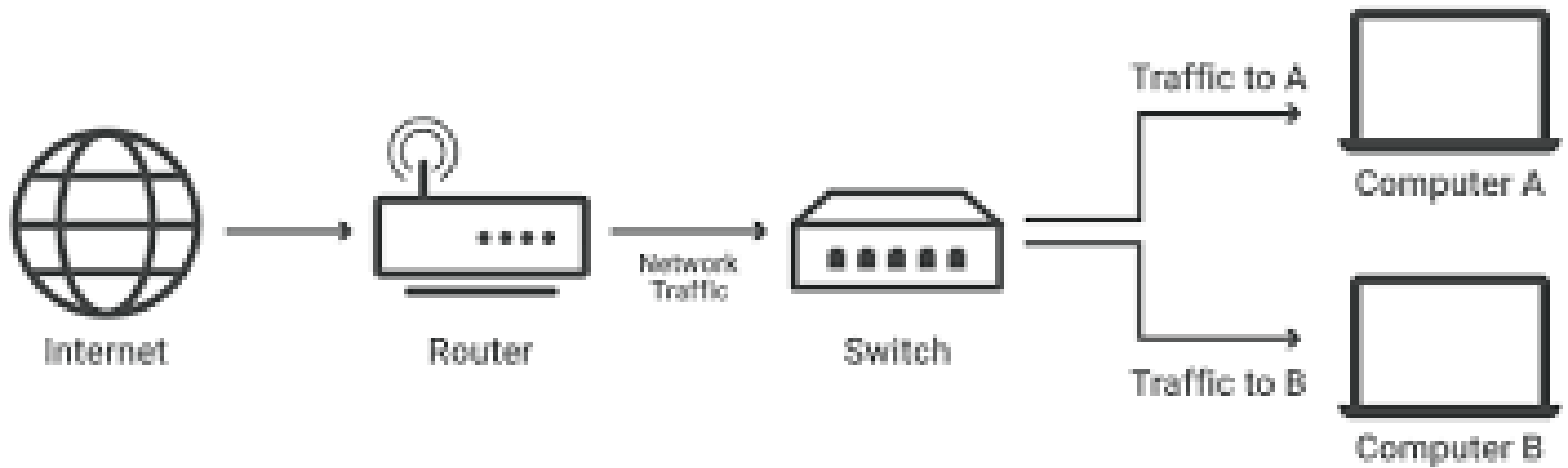






SWITCH NETWORK SETUP

- The image shows six PCs (A-F) connected to a Switch.
- PC A is sending a message specifically addressed to PC F.
- The Switch only forwards the message directly to PC F, not to all other PCs.
- This is because a Switch is intelligent — it reads the destination MAC address and sends data only to the intended device.
- More secure than hub.

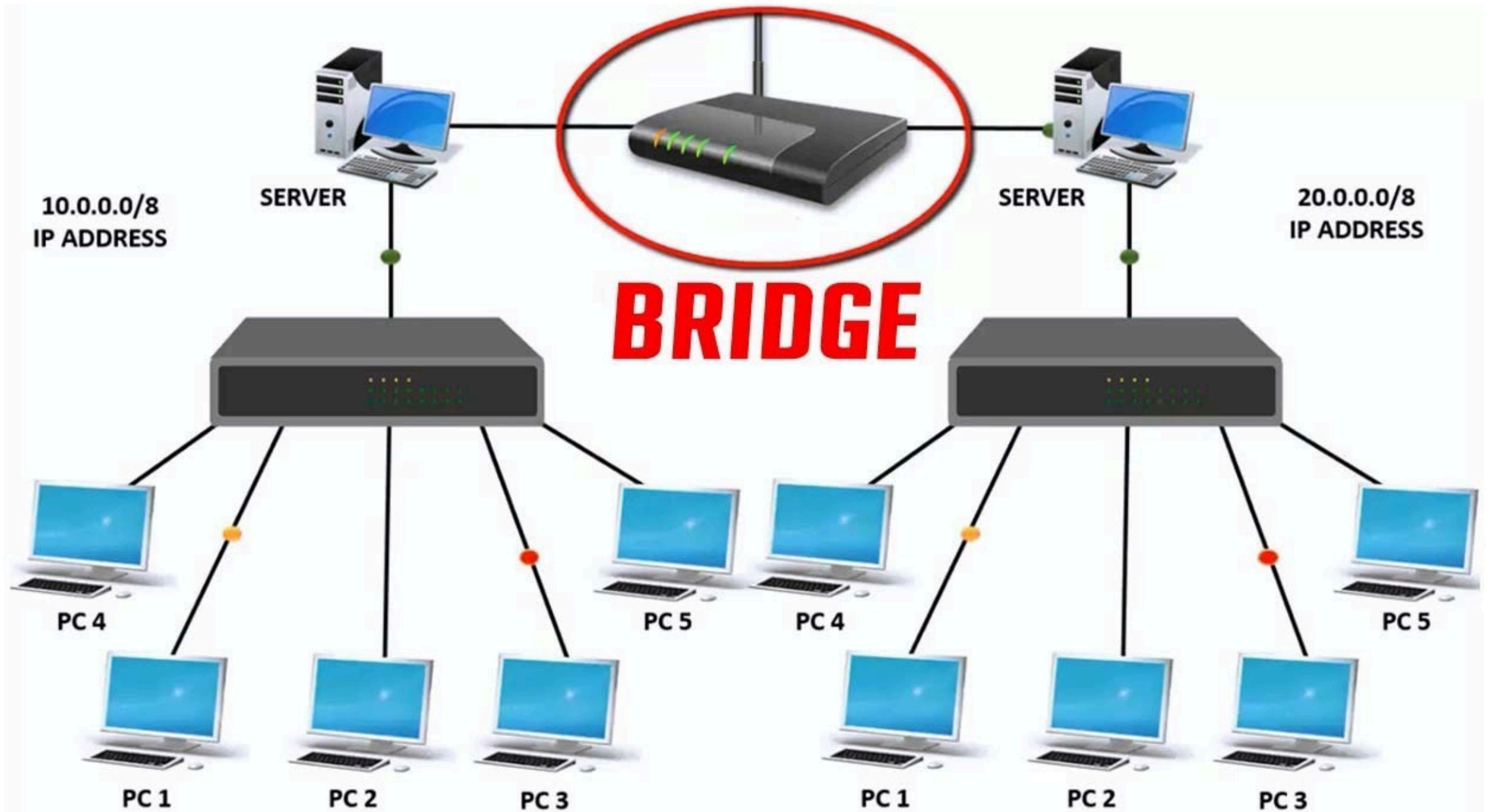


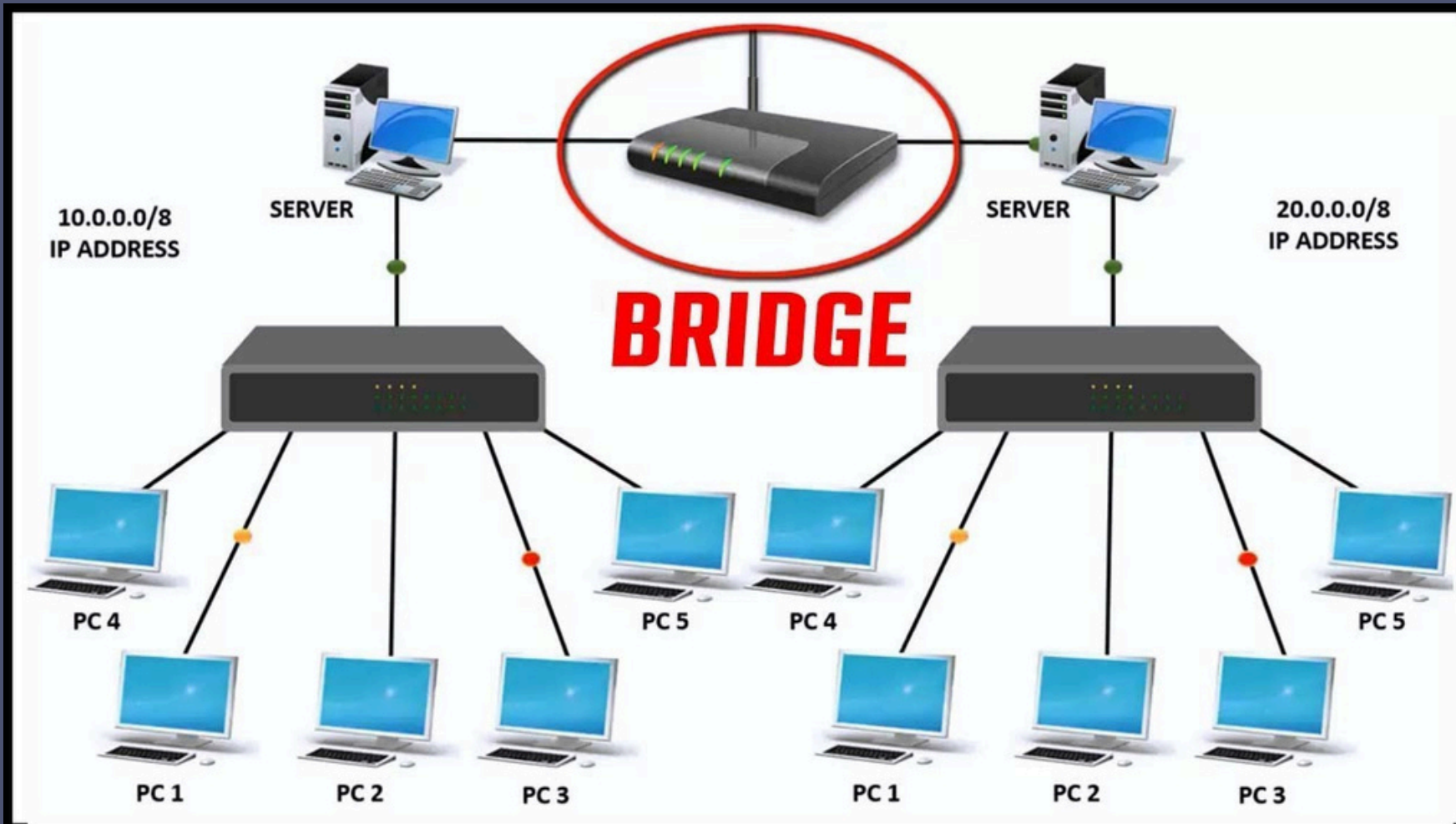
BRIDGES

BRIDGES DEVICE

- are device that connects one LAN to another LAN that uses the same protocol (communication rule).
- They are often used to connect together different parts of a LAN so that they can function as a single LAN.







BRIDGE NETWORK SETUP

- The diagram shows two different LANs (Local Area Networks).
- Left LAN uses the 10.0.0.0/8 IP address range.
- Right LAN uses the 20.0.0.0/8 IP address range.
- Each LAN has its own server and multiple PCs connected.
- The Bridge (in the red circle) connects these two separate networks, allowing them to communicate.

ROUTER



- Are used to route data packets from one network to another network, based on IP Addresses
- Each router has its own IP Address and used to join a LAN to the internet.

WIFI AND BLUETOOTH

- Sends and received radio waves in several different frequencies bands - 2.4 GHz and 5 GHz
- Offers fast data transfer than and better security than Bluetooth
- Sends and receives radio waves in a band of 79 different frequencies

FEATURE	BLUETOOTH	WIFI
Transmission frequency	2.45 GHz	2.4, 3.6, 5.0 GHz
Data transfer rate	25 Mbits/seconds	250 Mbits/second
Maximum effective range (meter)	30 METERS	100 METERS as long as there is barrier like walls
Maximum number of device	upto 7	depends on the capacity of the router

CLOUD COMPUTING (STORAGE)

- It is a method of data storage where data is stored on remote servers, there may be thousands of servers in many different locations

EXAMPLE:

- **Google Drive** because it allows users to store files, such as documents, photos, and videos, on Google's servers instead of on their local computers.
- These files can be accessed from any device with an internet connection, allowing users to share, edit, and collaborate on documents in real-time with others.
- The storage and processing are handled by Google's cloud infrastructure, rather than the user's local machine.

EXAMPLE:

- Google Drive
- GitHub
- Dropbox
- Microsoft OneDrive
- Apple iCloud
- Google Cloud Storage
- IBM Cloud Object Storage

COMMON NETWORK ENVIRONMENT

Intranet - private networks used within organizations

Extranet - extend the functionality of an intranet to include external users, such as partners, suppliers, or clients, providing controlled access to specific parts of the network

Internet - is a global network that connects millions people, allowing for the exchange of information and access to services worldwide

TYPES NETWORK

LOCAL AREA NETWORK (LAN)

WIRELESS AREA NETWORK (WLAN)

WIDE AREA NETWORK (WAN)

LAN (Local Area Network) is usually in one building or near areas. Limited connection only

Consist of computers and devices (printers) which will be connected to hubs or switches.

Functions of LANs:

- Allow sharing
- Easy communication between users of the LAN (Messaging directly from the network)
- admin can monitor the activity
- Can reach 10 m to 1000 m only

LOCAL AREA NETWORK

WLAN (Wireless Local Area Network) similar to LAN, but no wires or cables connected but using signals. Devices known as access point (APs)

Functions of LANs:

- Without using cables
- allow sharing
- communication

WIRELESS LAN

WLAN (Wired Local Area Network) refers to LAN, but uses physical cables (usually Ethernet) to connect devices.

Functions of LANs:

- using cables
- allow sharing
- communication

**WIRED
LAN**

WAN (Wide Area Network) are used where computers or networks are situated a long distance from each other.

Example:

- If the number of LAN are joined together using router, they it can form a WAN.
- Can reach 100 km to over 1000 km

WIDE AREA NETWORK

NETWORKS ISSUES AND COMMUNICATION

USE OF PASSWORD WHEN ACCESSING THE INTERNET

When accessing your email account

When carrying out online banking

When accessing your social media accounts

OTHER AUTHENTICATION METHODS

Zero Login

Biometrics

Magnetic Stripes

Smart Cards

Physical tokens

Electronic tokens

OTHER AUTHENTICATION METHODS

Zero Login - ID, Face Scan, others

Biometrics - fingerprint, ID, Face Scan, others

Magnetic Stripes - swipe the card to login

Smart Cards - contactless card to login/ scan
the chip from your card

Physical tokens and Electronic tokens

- OTP authentication through SmartPhone

FOR SECURITY OF THE ACCOUNTS

Use Anti - malware software

- anti virus software, firewall

Use a strong password

- use the combination of letters, numbers and special characters like *&+%.>>

TYPES OF ELECTRONIC CONFERENCING

Video Conferencing

Audio Conferencing

Web Conferencing

VIDEO CONFERENCING



- It is a communication method that uses both video and sound. By the use of Webcam, microphone software driver from the device

AUDIO CONFERENCING



- It is a communication between people using audio (sound) equipment. You also use telephone of smartphone

WEB CONFERENCING



- (often referred to as webinar or webcasts) uses the internet to permit conferencing to take place.